

Advanced Linear Algebra: Foundations to Frontiers

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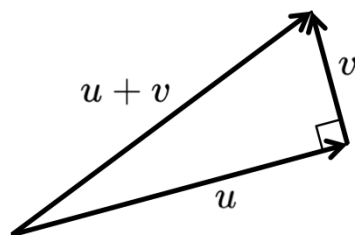
2.2.1 Orthogonal vectors ¶

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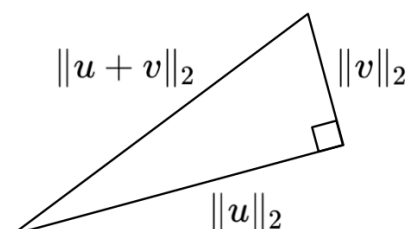
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At some point in your education you were told that vectors are orthogonal (perpendicular) if and only if their dot product (inner product) equals zero. Let's review where this comes from. Given two vectors $u, v \in \mathbb{R}^m$, those two vectors, and their sum all exist in the same two dimensional (2D) subspace. So, they can be visualized as



where the page on which they are drawn is that 2D subspace. Now, if they are, as drawn, perpendicular and we consider the lengths of the sides of the triangle that they define



then we can employ the first theorem you were probably ever exposed to, the Pythagorean Theorem, to find that

$$\|u\|_2^2 + \|v\|_2^2 = \|u + v\|_2^2.$$

Using what we know about the relation between the two norm and the dot product, we find that

$$\begin{aligned} u^T u + v^T v &= (u + v)^T (u + v) \\ \Leftrightarrow &< \text{multiply out} > \\ u^T u + v^T v &= u^T u + u^T v + v^T u + v^T v \\ \Leftrightarrow &< u^T v = v^T u \text{ if } u \text{ and } v \text{ are real-valued} > \\ u^T u + v^T v &= u^T u + 2u^T v + v^T v \\ \Leftrightarrow &< \text{delete common terms} > \\ 0 &= 2u^T v \end{aligned}$$

so that we can conclude that $u^T v = 0$.

While we already encountered the notation $x^H x$ as an alternative way of expressing the length of a vector, $\|x\|_2 = \sqrt{x^H x}$, we have not formally defined the inner product (dot product), for complex-valued vectors:

Definition 2.2.1.1. Dot product (Inner product). Given $x, y \in \mathbb{C}^m$ their dot product (inner product) is defined as

$$x^H y = \overline{x}^T y = \overline{x^T y} = \overline{\chi_0} \psi_0 + \overline{\chi_1} \psi_1 + \cdots + \overline{\chi_{m-1}} \psi_{m-1} = \sum_{i=0}^{m-1} \overline{\chi_i} \psi_i.$$

The notation x^H is short for $\overline{x}^T$, where $\overline{x}$ equals the vector x with all its entries conjugated. So,

$$x^H y$$

= < expose the elements of the vectors >

$$\begin{pmatrix} \chi_0 \\ \vdots \\ \chi_{m-1} \end{pmatrix}^H \begin{pmatrix} \psi_0 \\ \vdots \\ \psi_{m-1} \end{pmatrix}$$

= < $x^H = \bar{x}^T$ >

$$\begin{pmatrix} \chi_0 \\ \vdots \\ \chi_{m-1} \end{pmatrix}^T \begin{pmatrix} \psi_0 \\ \vdots \\ \psi_{m-1} \end{pmatrix}$$

= < conjugate the elements of x >

$$\begin{pmatrix} \bar{\chi}_0 \\ \vdots \\ \bar{\chi}_{m-1} \end{pmatrix}^T \begin{pmatrix} \psi_0 \\ \vdots \\ \psi_{m-1} \end{pmatrix}$$

= < view x as a $m \times 1$ matrix and transpose >

$$\left(\begin{array}{c|c|c} \bar{\chi}_0 & \cdots & \bar{\chi}_{m-1} \end{array} \right) \begin{pmatrix} \psi_0 \\ \vdots \\ \psi_{m-1} \end{pmatrix}$$

= < view x^H as a matrix and perform matrix-vector multiplication >

$$\sum_{i=0}^{m-1} \bar{\chi}_i \psi_i.$$

Homework 2.2.1.1. Let $x, y \in \mathbb{C}^m$.

ALWAYS/SOMETIMES/NEVER: $\overline{x^H y} = y^H x$.

▼ Answer

▼ Solution

$$\overline{x^H y} = \overline{\sum_{i=0}^{m-1} \bar{\chi}_i \psi_i} = \sum_{i=0}^{m-1} \overline{\bar{\chi}_i \psi_i} = \sum_{i=0}^{m-1} \psi_i \chi_i = y^H x.$$

ALWAYS

Now prove it!

Homework 2.2.1.2. Let $x \in \mathbb{C}^m$.

ALWAYS/SOMETIMES/NEVER: $x^H x$ is real-valued.

▼ Answer

▼ Solution

By the last homework,

$$\overline{x^H x} = x^H x,$$

A complex number is equal to its conjugate only if it is real-valued.

ALWAYS

Now prove it!

The following defines orthogonality of two vectors with complex-valued elements:

Definition 2.2.1.2. Orthogonal vectors. Let $x, y \in \mathbb{C}^m$. These vectors are said to be orthogonal (perpendicular) iff $x^H y = 0$.